DBMS - **SQL Query Implementation for E-Commerce Database**

Snapdeal_ecommerce_project

1.SELECT, WHERE, ORDER BY, and DISTINCT queries.

SELECT Query

The SELECT statement is used to retrieve data from a table.

Display all products:

```
SQL> SELECT * FROM Product;

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
CATEGORY_ID      PRICE      STOCK
----------- ---------- ----------
DESCRIPTION
--------------------------------------------------------------------------------
        23
Nike Running Shoes
          1       4499         15
Lightweight running shoes


PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
CATEGORY_ID      PRICE      STOCK
----------- ---------- ----------
DESCRIPTION
--------------------------------------------------------------------------------
        24
Adidas Casual Shoes
          1       3599         20
Comfortable casual shoes
```

Display Specific Product Details:

```
SQL> SELECT Product_ID, Product_Name, Price
  2  FROM Product;

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
     PRICE
----------
        23
Nike Running Shoes
      4499

        24
Adidas Casual Shoes
      3599

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
     PRICE
----------

        25
Puma Sports Shoes
      2999

        26
Atomic Habits
```

WHERE Clause

The WHERE clause is used to filter records based on a condition.

Search Products Based on Price (WHERE):

```
SQL> SELECT *
  2  FROM Product
  3  WHERE Price < 1000;

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
CATEGORY_ID      PRICE      STOCK
----------- ---------- ----------
DESCRIPTION
--------------------------------------------------------------------------------
        26
Atomic Habits
         2        499         30
Self improvement book
```

Search Products Based on Category:

```
SQL> SELECT *
  2  FROM Product
  3  WHERE Category_ID = 1;

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
CATEGORY_ID      PRICE      STOCK
----------- ---------- ----------
DESCRIPTION
--------------------------------------------------------------------------------
        23
Nike Running Shoes
          1       4499         15
Lightweight running shoes


PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
CATEGORY_ID      PRICE      STOCK
----------- ---------- ----------
DESCRIPTION
--------------------------------------------------------------------------------
        24
Adidas Casual Shoes
          1       3599         20
Comfortable casual shoes
```

Display Out-of-Stock Products:

```
SQL> SELECT *
  2  FROM Product
  3  WHERE Stock = 0;

no rows selected

SQL>
```

ORDER BY

The ORDER BY clause is used to sort the query results.

Display Products in Ascending Price Order (ORDER BY):

```
SQL> SELECT Product_ID, Product_Name, Price
  2  FROM Product
  3  ORDER BY Price ASC;

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
     PRICE
----------
        33
Tata Salt 1kg
        30

        34
Bru Instant Coffee 100g
       275
```

Display Products in Descending Price Order:

```
SQL> SELECT Product_ID, Product_Name, Price
  2  FROM Product
  3  ORDER BY Price DESC;

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
     PRICE
----------
        23
Nike Running Shoes
      4499

        24
Adidas Casual Shoes
      3599

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
     PRICE
----------

        25
Puma Sports Shoes
      2999

        36
Cosco Cricket Bat
```

DISTINCT

The DISTINCT keyword is used to display unique values.

```
SQL> SELECT DISTINCT Category_ID
  2  FROM Product;

CATEGORY_ID
-----------
          1
          2
          3
          4
          5
```

Search Products by Name:

```
SQL> SELECT *
  2  FROM Product
  3  WHERE Product_Name LIKE '%Shoes%';

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
CATEGORY_ID      PRICE      STOCK
----------- ---------- ----------
DESCRIPTION
--------------------------------------------------------------------------------
        23
Nike Running Shoes
          1       4499         15
Lightweight running shoes


PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
CATEGORY_ID      PRICE      STOCK
----------- ---------- ----------
DESCRIPTION
--------------------------------------------------------------------------------
        24
Adidas Casual Shoes
          1       3599         20
Comfortable casual shoes
```

2.To search products based on price, category, and availability

```
SQL> SELECT *
  2  FROM Product
  3  WHERE Price < 1000;

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
CATEGORY_ID      PRICE      STOCK
----------- ---------- ----------
DESCRIPTION
--------------------------------------------------------------------------------
       26
Atomic Habits
         2        499         30
Self improvement book
```

Search available products:

```
SQL> SELECT *
  2  FROM Product
  3  WHERE Stock > 0;

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------
CATEGORY_ID      PRICE      STOCK
----------- ---------- ----------
DESCRIPTION
--------------------------------------------------------
        23
Nike Running Shoes
          1       4499         15
Lightweight running shoes
```

3. To retrieve customer and product information from the database.

```
SQL> SELECT
  2      p.Product_ID,
  3      p.Product_Name,
  4      c.Category_Name,
  5      p.Price,
  6      p.Stock
  7  FROM Product p
  8  JOIN Category c
  9  ON p.Category_ID = c.Category_ID;

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
CATEGORY_NAME                                          PRICE      STOCK
-------------------------------------------------- ---------- ----------
        23
Nike Running Shoes
Footwear                                                4499         15

        24
Adidas Casual Shoes
Footwear                                                3599         20
```

4. To apply filtering conditions to retrieve specific product details.

Products priced between ₹300 and ₹1000:

```
SQL> SELECT *
  2  FROM Product
  3  WHERE Price BETWEEN 300 AND 1000;

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------
CATEGORY_ID      PRICE      STOCK
----------- ---------- ----------
DESCRIPTION
--------------------------------------------------------
        26
Atomic Habits
          2        499         30
Self improvement book
```

Products with stock greater than 20:

```
SQL> SELECT Product_ID, Product_Name, Stock
  2  FROM Product
  3  WHERE Stock > 20;

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------
     STOCK
----------
        26
Atomic Habits
        30

        27
Python Programming
        25
```

5. To generate basic business reports using SQL queries.

Total number of products:

```
SQL> SELECT COUNT(*) AS Total_Products
  2  FROM Product;

TOTAL_PRODUCTS
--------------
            14
```

Minimum and maximum product prices:

```
SQL> SELECT
  2      MIN(Price) AS Lowest_Price,
  3      MAX(Price) AS Highest_Price
  4  FROM Product;

LOWEST_PRICE HIGHEST_PRICE
------------ -------------
          30          4499
```

Average product price:

```
SQL> SELECT AVG(Price) AS Average_Price
  2  FROM Product;

AVERAGE_PRICE
-------------
   1217.35714
```

Count products in each category:

```
SQL> SELECT
  2      c.Category_Name,
  3      COUNT(p.Product_ID) AS Total_Products
  4  FROM Category c
  5  LEFT JOIN Product p
  6  ON c.Category_ID = p.Category_ID
  7  GROUP BY c.Category_Name
  8  ORDER BY c.Category_Name;

CATEGORY_NAME                                      TOTAL_PRODUCTS
-------------------------------------------------- --------------
Beauty                                                          3
Books                                                           3
Footwear                                                        3
Grocery                                                         3
Sports                                                          2
```

Conclusion

These SQL queries help search products, retrieve product and category information, filter records, and generate basic business reports for the Snapdeal e-commerce database.